Code run-book for Claude Code: bc-grade-opendata v1.0

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RUNBOOK — taking v0.1 to a submission-ready paper with Claude Code

This runbook is written for a Claude Code session (or a human analyst) working on a branch of this repository. Each step has an **action**, the **files it touches** and an **acceptance check**. Work top to bottom; tick the status table as you go and commit after each step.

```
git checkout -b submission-run
claude                      # then: "Read CLAUDE.md and RUNBOOK.md, then do step R1"
```

Status at v1.0 (2026-10-18, submission build)

Area	State
R1 data + guards	Done — checksums unchanged since v0.1; 12 tests pass (2 new: MI pooling, Duke in-situ/components)
R2 exact reproduction of v0.1	Done — `numbers.tex` identical after a clean re-download and re-run
R3 submission resampling (10 repeats, 2000 bootstraps)	Done
R4 multiple imputation (m = 20, pooled predictions) + complete case	Done — `e2mi`, `tab_mi`
R5 TabPFN	**Blocked** — weights gated on Hugging Face (login + licence) and huggingface.co unreachable from the build sandbox; code path in `make_model("tabpfn")`, runs automatically when weights are cached (`optional_models_status.json`)
R6 more Saudi/Gulf anchors	Partly done — Riyadh (Durrani 2025), Najran, Madinah added; pooled random-effects anchor; age-share sensitivity; bootstrap CIs. **All new values are abstract/snippet-level** (journal sites unreachable) → V2 must re-read the tables
R7 sample size (Riley/Pate multinomial)	Done — `e10`; criteria **not met** for the G1-vs-G2 sub-model (reported in the paper)
R8–R9 image-level arm (KAU-BCMD, CDD-CESM, BUSI)	**Blocked** — Kaggle/TCIA hosts unreachable from the build sandbox
R10 Duke-Breast-Cancer-MRI	Done — clinical + 529 radiomics from a checksummed GitHub mirror of the TCIA XLSX; Duke is a second full-feature external cohort, in LODO, leakage audit (component scores) and E9 radiomics fusion
R11 manuscript	Done — rebuilt: Figure 1 framework (TikZ), pgfplots charts from `generated/data/*.csv`, new Introduction/Discussion, abstract
V1 citations	Done via search-engine records (79 entries; `paper/refs_annotated.bib` keeps provenance notes). One DOI-resolver pass from an unrestricted machine still advised
V2 anchor double-read	**To do (human)**
V3 TRIPOD+AI full checklist	Done (Supplementary Table) — STROBE checklist still to add
V4 independent clean re-run	To do on a second machine
G1 GitHub + Zenodo	Package ready (`.github/workflows/ci.yml` added); repository creation + DOI = PI
Author list, affiliations, funding, CRediT, PPI statement	To do PI

Headline v1.0 results (from `paper/generated/numbers.tex`): internal macro-AUC 0.72–0.73 for every learned model, XGBoost − LR = +0.001 (95% CI −0.006 to +0.008); external full-feature macro-AUC 0.69–0.71 (MSK-2018), 0.70–0.72 (Duke); grade-3 calibration slope ≈ 1 externally but intercept +0.99 (MSK) and −1.02 (Duke), both fixed by intercept update; MI vs single imputation $\Delta \leq 0.003$; leakage inflation up to +0.30; Duke radiomics fusion −0.045 vs clinical; Riley sample size not met (4477 required vs 2386); Saudi case-mix raises each cohort's grade-3 share by ≈ +10 points, and a METABRIC model would predict 59% grade 3 vs 36.7% published (Dammam).

Headline v0.1 results (superseded; see `paper/generated/numbers.tex` for v1.0) (all from `paper/generated/numbers.tex`): internal macro-AUC ≈ 0.72 for every learned model (logistic

regression = XGBoost); external macro-AUC 0.64–0.80; grade-3 calibration slope ≈ 1.03 in MSK-2018 but intercept ≈ +0.99 (baseline-risk shift, fixed by intercept update); 7/7 SHAP directions plausible; leakage inflates macro-AUC by up to +0.27; Saudi re-weighting implies ≈ 61% grade 3 vs 36.7% published → Western models would over-predict grade 3 in Saudi patients.

R0 — Environment (15 min)

Action. `python3 -m venv .venv && . .venv/bin/activate && pip install -r requirements.txt`. LaTeX: TeX Live with `charter`, `natbib`, `booktabs`, `enumitem`, `hyperref`, `latexmk`. **Accept.** `python -c "import sklearn, xgboost, lightgbm, shap, statsmodels, imblearn"` succeeds.

R1 — Data + guards (5 min)

Action. `make data test`. **Accept.** 10 tests pass. `git diff data/raw/MANIFEST.json` is empty; if checksums changed, cBioPortal updated a study — record the date in `paper/main.tex` Data availability and re-run everything.

R2 — Reproduce v0.1 exactly (≈ 15 min, 2 cores)

Action. `make experiments report paper`. **Accept.** `git diff paper/generated/numbers.tex` shows no change (the seed is fixed). If numbers move by more than ±0.002, compare package versions with `results/run_*.json` before continuing.

R3 — Submission-grade resampling

Action. In `configs/experiment.yaml` set `cv.repeats: 10`, `bootstrap: 2000`. Run `make experiments report paper`. **Accept.** Methods text updates automatically through `\Repeats` / `\Boot`. Macro-AUCs change < 0.005.

R4 — Multiple imputation with pooled predictions

Why. Reviewers will ask; v0.1 uses single in-fold chained-equation imputation. **Action.** In `src/bcgrade/models.py` add `MIClassifier(m=20)`: inside `fit`, create m `IterativeImputer(sample_posterior=Tr random_state=seed+i)` → scaler → clone(model) pipelines; in `predict_proba`, average the m probability matrices. Register as model option `mi=True`; add a test that predictions are averaged over m distinct imputations. Add an experiment `e2mi` for `mlr` and `xgb` only. Add a complete-case sensitivity row. Report in a new supplementary table `tab_mi.tex`. **Accept.** MI and single-imputation macro-AUC differ by < 0.01, or the paper discusses why not.

R5 — Tabular foundation model (TabPFN)

Action. `pip install tabpfn` (needs model weights from Hugging Face on first use). Add `"tabpfn"` to `make_model` (no tuning grid; `TabPFNClassifier(random_state=seed)`), include in E2/E3/LODO. Cite `hollmann2025tabpfn` (already in refs.bib). **Accept.** Runs within the CV harness without leakage (it sits after impute/scale in the Pipeline). Expectation from v0.1: it will not beat regression by more than the CI; report either way.

R6 — Strengthen the Saudi / Gulf anchor (key for sponsors)

Action. 1. Find further published Saudi/Gulf clinicopathological series reporting grade, ER/PR/HER2, histology and age (Saudi Medical Journal, Annals of Saudi Medicine, J Clin Med, Cureus, BMC Cancer; KFSH&RC Riyadh, KAMC Riyadh/Jeddah, KAUH Jeddah; also UAE, Kuwait, Oman, Qatar). Transcribe Table 1 counts into `configs/anchors.yaml` (one block per study, with page and table numbers). 2. Generalise `e7` to loop over all anchors and add a random-effects

pooled Saudi grade distribution. 3. Add a "joint" sensitivity: rake on ER × HER2 jointly if any study reports the cross-tab. 4. Add Saudi Cancer Registry figures used in the Introduction (Basudan 2022 already; add the most recent SCR report). **Accept.** Every anchor value has a citation + table/page reference in a YAML comment and is double-checked by a second reader (V2).

R7 — Sample-size justification

Action. Use `pmsampsize` (R) or a Python port for a multinomial outcome (Riley et al. BMJ 2020; Pate et al. for multinomial). Inputs: 10 parameters, smallest class prevalence 0.089, anticipated Cox–Snell R^2 from v0.1 (compute from METABRIC logistic model). Write the output into `results/sample_size.json` and a sentence in Methods via a macro. **Accept.** Required n ≤ 2386, or the paper says it is not met.

R8 — Imaging arm B1: KAU-BCMD (Saudi mammography)

Action. 1. `pip install kaggle`; place `~/.kaggle/kaggle.json`; `python scripts/download_imaging.py kau_bcmd`. 2. Build `src/bcgrade/imaging/kau.py`: index images → patient/case id → BI-RADS label; record the version (the Kaggle folders hold BI-RADS 1, 3, 4, 5; the paper describes 1–5 — document what you downloaded). 3. Task: BI-RADS 1 vs 3 vs 4–5 (or benign-appearing vs suspicious), **StratifiedGroupKFold by case**. 4. Models: frozen ImageNet ViT/ConvNeXt or a mammography foundation model → embeddings → logistic regression; then fine-tuned EfficientNet-B0 on a GPU. Same metrics/calibration code. 5. Leakage demo: image-level vs case-level split (report like E4-d). **Accept.** No case appears in two folds (unit test). Results in a new section "Imaging arm".

R9 — Imaging arm B2: CDD-CESM (Egypt) and BUSI (Egypt)

Action. `python scripts/download_imaging.py cdd_cesm --manifest-only`, then full download; BUSI via Kaggle. Tasks: pathology benign vs malignant (CDD-CESM, BUSI) and BI-RADS (CDD-CESM). Cross-site check: train on CDD-CESM low-energy images, test on KAU-BCMD for a "suspicious vs not" label mapping (document the mapping table in `configs/`). **Accept.** Patient-grouped CV; external cross-site AUC reported with CI.

R10 — Duke-Breast-Cancer-MRI clinical + radiomics (restores BI-RADS-era variables)

Why. It is the only open cohort linking Nottingham grade to mammographic calcification, density and MRI radiomics (922 patients; CC BY-NC 4.0). **Action.** Download the clinical XLSX and the 529-feature radiomics XLSX from TCIA (no image download needed). Add `duke()` to `harmonise.py` (map Nottingham grade; drop tubule/nuclear/mitotic scores — leakage; code calcification and density). Add Duke as (a) external validation cohort for the full CDM model, (b) a clinical + radiomics fusion model (E9), compared with the published radiomics benchmark (validation AUC 0.722 for grade 3 vs 1–2; cite after V1 verification). **Accept.** New external row for Duke; licence note (non-commercial) added to Data availability.

R11 — Manuscript update

Action. Re-run `make report paper`. Rewrite only prose; never hard-code a number (CLAUDE.md rule 1). Update the abstract (≤ 300 words for most journals) and the TRIPOD+AI table. **Accept.** `grep -nE "[0-9]\.[0-9]{2,}" paper/main.tex paper/sections/*.tex` shows only published anchor figures (Saudi/registry) and software versions, no results.

V1 — Citation verification

Check every `refs.bib` entry with a DOI resolver; fix all `VERIFY` notes (wang2022duke, kim2025grade, kussaibi2021saudi DOI, mbcproject2022). Remove uncited entries.

V2 — Anchor verification

Second reader re-transcribes `configs/anchors.yaml` from the source PDFs blind, then diff.

V3 — Reporting checklists

Complete the full TRIPOD+AI (27 items) and STROBE checklists as supplementary PDFs; replace the abbreviated Table S5.

V4 — Independent re-run

On a clean machine: `git clone … && make all`; `numbers.tex` must match the committed version.

G1 — GitHub + Zenodo release

1. Create the GitHub repository (public), push `main` (data and `results/*.npz` are git-ignored).
2. Enable the Zenodo–GitHub integration, tag `v1.0.0`, cite the DOI in `paper/main.tex` and `CITATION.cff`.
3. Add a GitHub Action running `make data test` on push (CPU, < 5 min).

Journal targeting note

Q1 options that publish open-data prediction studies with TRIPOD+AI: *Breast Cancer Research and Treatment*, *The Breast*, *JCO Clinical Cancer Informatics*, *npj Breast Cancer* (needs imaging/genomic angle — R8–R10), *Diagnostic and Prognostic Research* (methods angle: the leakage audit). Decide after R10.

CLAUDE.md — working rules for Claude Code in this repository

This repo is the reproducibility package for the manuscript in `paper/main.tex` (open-data breast cancer histological-grade prediction). Start every session by reading `RUNBOOK.md` and the "Status" table at its top.

Non-negotiable rules

1. **No number is typed by hand into the paper.** Every figure, table and in-text number comes from `results/` → `experiments/report.py` → `paper/generated/` (`numbers.tex` macros, `tab_*.tex`, `fig_*.pdf`). If you change an analysis, re-run `make experiments report paper` and read the diff of `numbers.tex`.
2. **Every learned step stays inside the sklearn/imblearn Pipeline** (array copy → imputation → scaling → optional SMOTE → model). Never fit an imputer, scaler, resampler or feature selector on data that includes a test fold. `tests/test_pipeline.py` guards this; add a test when you add a step.
3. **Leakage columns are never predictors** of grade: NPI (contains grade), Nottingham component scores (tubule/pleomorphism/mitotic), stage (contains T/N but also used for selection), PAM50/IntClust/Oncotype unless explicitly in a sensitivity analysis. Keep `test_leakage_columns_never_in_feature_sets` passing.
4. **In-situ disease is excluded, never relabelled.** ILC is never merged with IDC. Grade is invasive Nottingham/differentiation grade 1–3 only.
5. **One row per patient.** Any image or multi-sample source must be split with patient-grouped CV (`StratifiedGroupKFold`).
6. **Data are not redistributed.** `data/` is git-ignored. Download scripts + `data/raw/MANIFEST.json` (SHA-256) are the provenance record. Respect CC BY-NC terms (Duke MRI, BRACS). Never

use a source that needs a signed DUA (VinDr-Mammo, full EMBED) without the PI's written decision.

7. **Seeds**: `configs/experiment.yaml: seed`. Do not change the seed to improve a result.
8. Commit messages: imperative, one logical change each; never commit `data/` or large `results/*.npz`.

Layout

```
configs/    datasets.yaml (sources, licences, roles) · experiment.yaml (CV, seeds, raking targets) · a
scripts/    download_tabular.py (open cBioPortal + UCI mirror) · download_imaging.py (arm B, Kaggle/T
src/bcgrade/  harmonise.py (common data model) · describe.py (scale-appropriate stats) · models.py (la
experiments/  run.py (E1–E8) · report.py (tables/figures/macros)
tests/        pytest guards for harmonisation and leakage
paper/        main.tex · refs.bib · generated/ (never edit by hand)
```

Common commands

```
make data          # download + checksum
make test          # guards
make experiments   # E1–E8 (~15–25 min on 2 CPU cores at repeats=5)
make report paper  # regenerate tables/figures/macros and compile the PDF
python experiments/run.py e3 e5  # re-run selected stages (E5 and E7 depend on E2 + E3 outputs)
```